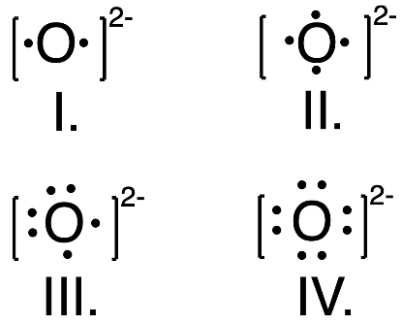
Test 4

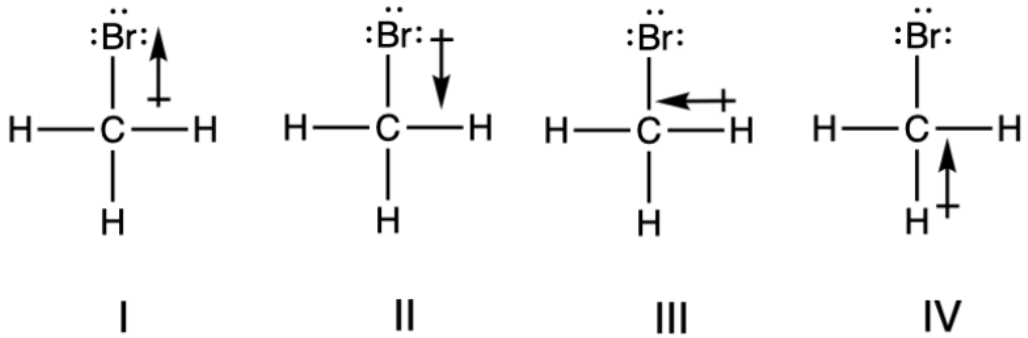
Name _____**Date**_____

Part I. Choose the most appropriate answer (2 points each)

1. Identify which of the following molecules has a double bond.
   a. H_2
   b. O_2
   c. Br_2
   d. HF
2. Select the correct Lewis symbol for Oxide ion.
   a. I
   b. II
   c. III
   d. IV

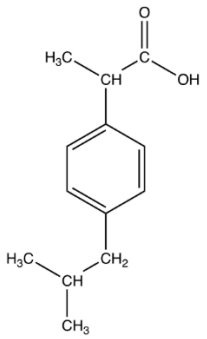


3. Identify the correct dipole for the C-Br bond in the Lewis structure of methyl bromide shown using a vector arrow.

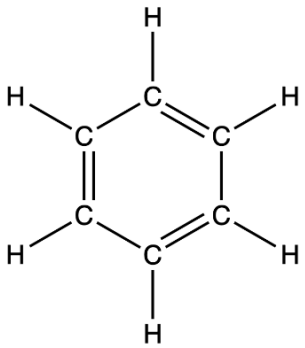


   a. I
   b. II
   c. III
   d. IV

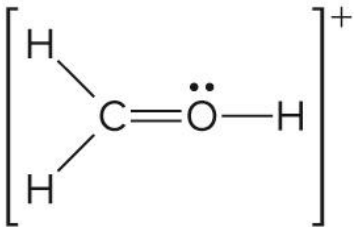
4. How many Pi bonds are present in the structure of ibuprofen shown?
   a. 2
   b. 4
   c. 5
   d. 3



5. What is the hybridization of all the carbon atoms in benzene (C_6H_6)



   a. Sp
   b. Sp^3
   c. Sp^2
   d. Sp^3d
6. A more electronegative atom
   a. Will have more attraction to the electrons in a chemical bond
   b. Is more likely to lose an electron
   c. Is less likely to form a chemical bond
   d. Is more likely to form ionic bond with another highly electronegative atom
7. What is the formal charge on the oxygen atom in the Lewis structure to the right?
   a. 0
   b. +1
   c. -1
   d. -2

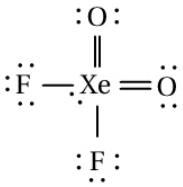


8. Nitrogen and hydrogen combine to form ammonia in the Haber process. Calculate (in kJ) the standard enthalpy change $\Delta H°$ for the reaction written below, using the bond energies given.
$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

Bond	N≡N	H—H	N—H
Bond energy (kJ/mol)	945	432	391

a. –969 kJ
b. –204 kJ
c. –105 kJ
d. 204 kJ

9. What is the molecular shape of XeO_2F_2 as predicted by the VSEPR theory?



a. square planar
b. tetrahedral
c. square pyramidal
d. see-saw

10. adding electrons to bonding molecular orbitals will
    a. increase the bond order
    b. decrease the bond order
    c. have no effect on the bond order
    d. weaken the bond

Part II. Calculations: To get full credit you must show all your work! (30 points)

1. **(16 points)** For each of the following molecules;
   - Draw a reasonable Lewis structure and resonance structure if indicated
   - Describe the molecular geometry and polarity, and
   - Give the hybridization on the central atom

Molecule	Lewis Structure(s)	Molecular geometry	Polarity	Central atom hybridization
AsH_3				
SNO^-				
SO_2				
AsF_5				

2. (**4 points)** Draw all the resonance structures of PO_4^{-3} (phosphate ion)

3. **(10 points)** Use the molecular orbital model to predict the bond order and magnetism of each of the following molecules:

(A) Ne_2

(B) P_2